Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Steel Plate Product

from

JSW Anjar

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
EPD registration number:	S-P-13786
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An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com

General information

Programme information

Programme:	The International EPD® System	
Program Operator	EPD International AB Box 210 60, SE-100 31 Stockholm, Sweden	Indian Regional Hub www.environdecindia.com
Website:	www.environdec.com	
E-mail:	info@environdec.com	

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): *PCR 2019:14 Construction products, version 1.3.2 Published on 2023.08.12. Based on CEN standard EN 15804. ISO standard ISO 21930 and CEN standard EN 15804 serve as the core PCR. UN CPC code 412*.
PCR review was conducted by: The Technical Committee of the International EPD®System. See https://www.environdec.com/about-us/the-international-epd-system-about-the-system for a list of members.
Life Cycle Assessment (LCA)
LCA accountability: *JSW Steel Limited.*
LCA and EPD developer: Dr. Rajesh Kumar Singh Thinkstep Sustainability Solutions Pvt. Ltd., a Sphera Company
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ EPD verification by individual verifier Third party verifier: Dr. Hüdai Kara, Metsims Sustainability Consulting, 4 Clear Water Place, Oxford OX2 7NL, UK Email: hudai.kara@metsims.com Approved by: The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No [Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programs, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs;cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterization factors); have equivalent content declarations; and be valid at the time of comparison.

For further information about comparability, see EN 15804 and ISO 14025.

Company information

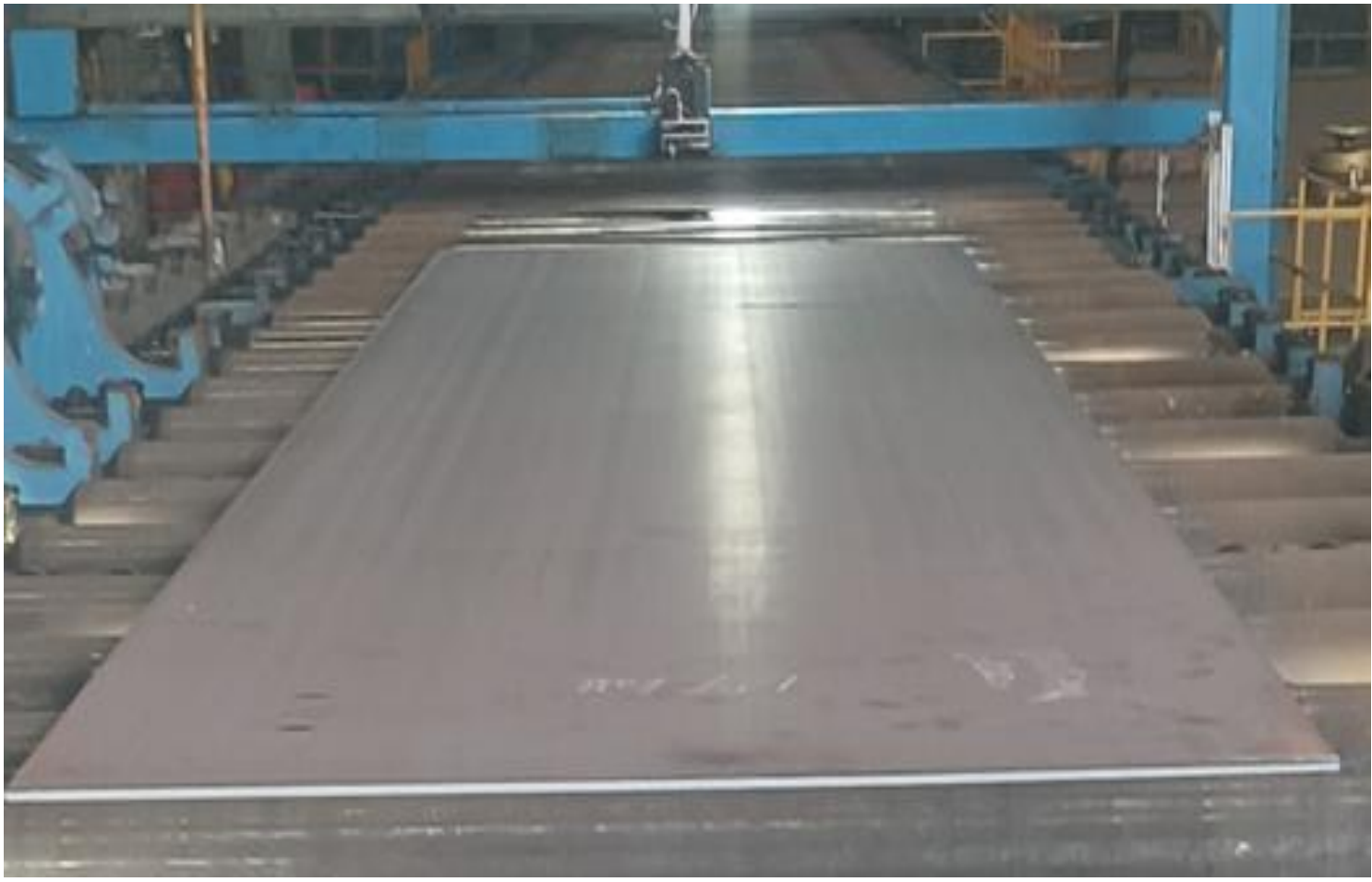
Owner of the EPD: JSW Steel Ltd, Anjar works.

Contact: hemal.bhagat@jsw.in; Tel: +91 9276819847

Description of the organization:
JSW Anjar offers High performance Steel Plate from the house of JSW group company. These Steel Plate are manufactured through a unique and technically advanced processes, resulting in optimized and higher strength, ductility, toughness and durability and versatility that are mainly used in Steel constructions.

Product-related or management system-related certifications:
All facilities at JSW Anjar are ISO 9001, ISO 14001 & ISO 45001 certified.

Name and location of production site(s):
JSW Steel Ltd, Anjar works
428H+M6X, Navhagam, Anjar, Gujarat 370110
Tel: 044 2361 1826



Product information

Product name: Steel Plate

Product identification: Steel Plate.

UN CPC code: 412.

Geographical scope: India.

Product description:

JSW Anjar manufactures steel plates using high-quality slabs sourced both domestically and internationally. Our plates are available in thicknesses ranging from 8 to 150mm, widths from 600 to 4500mm and length upto 28000 mm, catering to a wide range of applications. JSW Steel Plates that is specifically crafted with 100% grid electricity. Approximately 86% Slab quantity procured from Dolvi, Vijayanagar and Monnet (Raigarh) and remaining slab is imported from POSCO (Korea). Slab casting process takes place and then it is cut to produce Steel Plate at the Plate Mill. JSW leads the way in sustainable manufacturing.

Manufacturing process:

The steel Plate is, manufactured on the sites of Anjar, India. These are manufactured from hot rolled steel slab and subjected to advanced technology gives the uniform case and core ratio microstructure, it results in optimized strength and ductility combination.

Digitally Controlled Walking Beam Furnace from M/s Stein Heurtey is used for heating the slabs. High Pressure Primary De-Scaling System is used for removing the scales that have formed in the furnace. Reversible Mill from Mitsubishi Japan to roll the hot slabs into plates with desired dimensions.

Slab is brought from Dolvi, Vijayanagar, POSCO and Monnet plants which has been considered in the current assessment along with the transport.

Applications:

Steel Plate has a wide number of applications. Typical applications are:

1. **Oil and Gas Sector:** Steel A plenty of oil and gas companies employ steel plates in the pipelines to increase their strength and prevent corrosion, in addition to using steel for rig construction. Steel plates may be easily welded to meet the requirements of pipelines and can be used to repair pipes when necessary.
2. Infrastructure Development: Steel Plates are fundamental in infrastructure development projects, including pipelines, power plants, and railroads. Their adaptability and robustness enable the efficient execution of such projects, facilitating global infrastructure networks and fostering economic growth.
3. **Shipbuilding:** Steel plates, which are both lightweight and extremely durable, are often utilized in ship construction. Oil rigs and offshore structures also frequently make use of them. They are perfect for this since they are resistant to corrosion.

Product Range:

Steel Plate

Dimensions	Range (in mm)		
Grade	Thickness	Width	Length
S235, S275, S355 (JR, J0, J2, K2)	8-100	600-4500	3000-25000
S275, S355 (N, NL)	8-70	600-4500	3000-25000
S275, S355, S420, S460, S500 (M, ML)	6-25	600-4500	3000-25000
16Mo3, P235 GH, P265 GH, P295 GH, P355 GH	8-150	600-4500	3000-25000
Grade P275 NH, NL1 & NL2, and Grade P355 NH, NL1 & NL2	8-150	600-4500	3000-25000
C45	8-100	600-4500	3000-25000
E250, E300, E350, E410, E450, (A, BR, BO, C)	Up to 120	Up to 4500	Up to 25000
Grade- 1,2,3,4 & 5	Up to 110	Up to 4500	Up to 25000
R220, R260, R275, R355	8-150	Up to 4500	Up to 28000
GRADE 1,2 & 3	Up to 40	Up to 3500	Up to 15000
Grade - 55,60,70	8-150	Up to 4500	Up to 28000
A-36	Up to 120	Up to 4500	Up to 25000
Grade B, X42 to X-80	8-50	Up to 4500	Up to 25000

LCA information

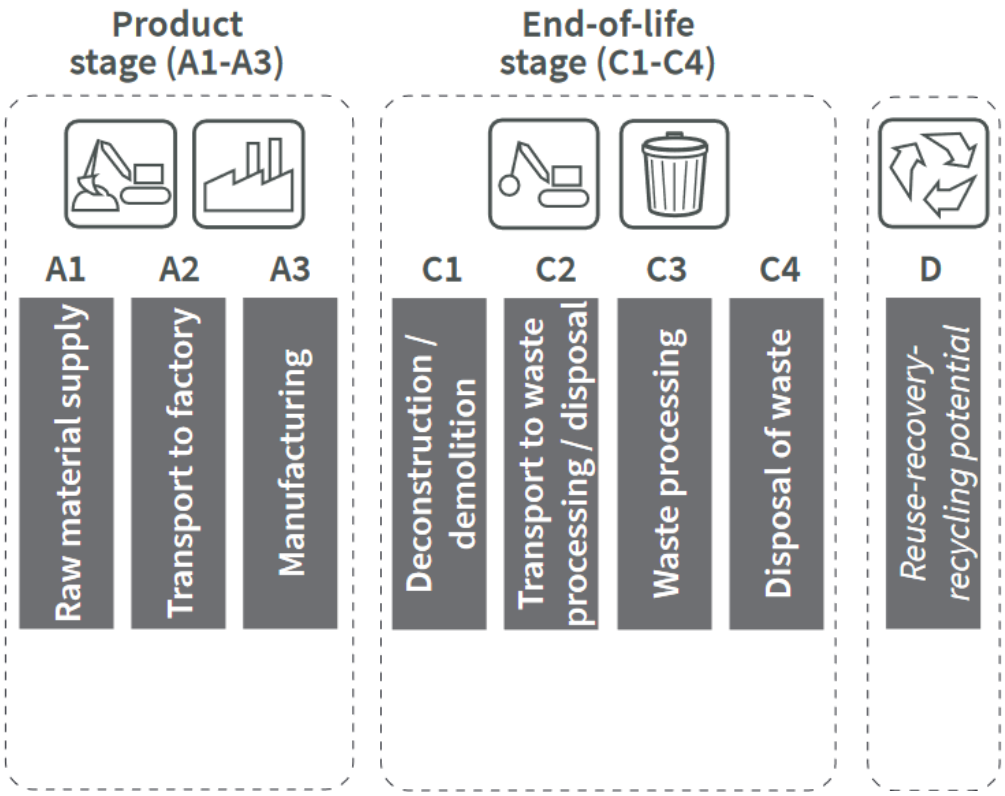
Functional unit / declared unit: 1 Tonne Steel Plate

Reference service life: Not applicable.

Time representativeness: The collection of foreground data refers to April 2022 to March 2023.

Database(s) and LCA software used:
The background data has been taken from the latest available Sphera Managed LCA Content 2023.2 and the LCA model was created using LCA Sphera for Experts software, version 10.7.

Description of system boundaries:
The system boundary: It includes cradle to gate stage with additional modules A1-A3, C1–C4, and module D.



- *Module A1 to A3:*

The product stage includes provision of all materials, products and energy, as well as waste processing up to the end-of waste state or disposal of final residues during the product stage. The electricity has been sourced from the grid. The emission factor for the GWP-GHG indicator (reference year 2019) is 1.06 kg CO_2eq./kWh.

Module C1 to C4:

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3) as well as the disposal (C4), if given. At EoL, the steel material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the steel material is 88% recycling and 12% landfill.

Category	Subcategory	Unit	Quantity
Collection process	Collected separately	kg	1000
	Collected with mixed construction waste	kg	0
Recovery	Reuse	kg	0
	Recycling	kg	880
	Landfill	kg	0
	Incineration	kg	0
	Incineration with energy recovery	kg	0
	Energy conversion efficiency rate	kg	0
Disposal	Material for final disposal	kg	120
Transport	Deconstruction site to scrap processing plant	km	0
	Scrap processing plant to site for end of waste	km	50

- *Module D:*

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials.

Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and loads of reuse or recycling potentials are grouped to module D.

Cut-off criteria:

The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, contributing less than 5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources:

Data quality is compliant with ISO 14025:2006. All primary data were collected for the year 2022-2023. All background data come from the Sphera Managed LCA Content 2023.2 databases.

Allocation:

With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049.

Price/economic allocation has been applied to manufacturing process (Main product- Steel plate & By product- Steel scrap).

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	NR	NR	NR	NR	NR	NR	NR	NR	NR	X	X	X	X	X
Geography	India			-	-	-	-	-	-	-	-	-	GLO	GLO	GLO	GLO	GLO
Specific data used	<90%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation –sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-

NR- Not reported. MNR- Module not declared.

Content information

MATERIAL Properties in Steel Plate

Product content	Weight, %	Post-consumer material, weight1	Biogenic material, weight
Steel		0%	0% and 0 kg C / kg
Chemical composition			
Iron	> 86		0
C	Max 0.60		0
S	Max 0.05		0
P	Max 0.05		0
Si	Max 0.60		0
Cr	Max 5.00		0
Ni	Max 2.00		0
Mn	Max 2.00		0
Cu	Max 1.50		0
V	Max 0.10		0
Nb	Max 0.10		0
Mo	Max1.50		0

Products do not contain any substances that can be included in “Candidate List of Substances of Very High Concern for Authorization” and raw materials used are not part of the EU REACH regulation.

Results of the environmental performance indicators

The environmental performance of the functional unit of one metric tonne of Steel Plate are reported below using the parameters and units as specified in PCR 2019:14 v1.3.2.

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Climate Change - total	kg CO_2 eq.	3.26E+03	0.00E+00	6.20E+00	0.00E+00	1.78E+00	-1.53E+03
Climate Change, fossil	kg CO_2 eq.	3.26E+03	0.00E+00	6.20E+00	0.00E+00	1.78E+00	-1.53E+03
Climate Change, biogenic	kg CO_2 eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Climate Change, (LULUC)	kg CO_2 eq.	6.13E-01	0.00E+00	8.21E-05	0.00E+00	5.60E-03	-2.03E-01
Ozone depletion	kg CFC -11 eq.	1.89E-06	0.00E+00	9.83E-14	0.00E+00	4.58E-12	2.05E-09
Acidification	Mole of H+ eq.	27.98	0.00E+00	5.44E-02	0.00E+00	1.28E-02	-3.74E+00
Eutrophication, freshwater	kg P eq.	8.45E-04	0.00E+00	1.33E-06	0.00E+00	3.63E-06	-3.56E-04
Eutrophication, marine	kg N eq.	4.79E+00	0.00E+00	2.57E-02	0.00E+00	3.30E-03	-6.01E-01
Eutrophication, terrestrial	Mole of N eq.	5.25E+01	0.00E+00	2.82E-01	0.00E+00	3.63E-02	-5.38E+00
Photochemical ozone formation, human health	kg NMVOC eq.	1.36E+01	0.00E+00	5.95E-02	0.00E+00	9.97E-03	-2.44E+00
Resource use, mineral and metals	kg Sb eq.	3.48E-04	0.00E+00	3.42E-08	0.00E+00	8.33E-08	-8.65E-03
Resource use, fossils	MJ	4.16E+04	0.00E+00	8.85E+01	0.00E+00	2.40E+01	-1.52E+04
Water use	m^3 world equiv.	1.26E+02	0.00E+00	7.43E-03	0.00E+00	1.98E-01	-1.03E+02
Acronyms	**Caption:** GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestric = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity.						

Additional mandatory and voluntary impact category indicators

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	3.25E+03	0.00E+00	6.18E+00	0.00E+00	1.78E+00	-1.52E+03
Additional voluntary indicators e.g. the voluntary indicators from EN 15804 or the global indicators according to ISO 21930:2017							

*Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator. We discourage the use of the results of modules A1-A3 without considering the results of module C.

Resource use indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	1.18E+03	0.00E+00	1.84E-01	0.00E+00	3.91E+00	6.00E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.18E+03	0.00E+00	1.84E-01	0.00E+00	3.91E+00	6.00E+02
PENRE	MJ	4.16E+04	0.00E+00	8.85E+01	0.00E+00	2.40E+01	-1.52E+04
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	4.16E+04	0.00E+00	8.85E+01	0.00E+00	2.40E+01	-1.52E+04
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	MJ	5.01E+00	0.00E+00	2.35E-04	0.00E+00	6.06E-03	-1.55E+02
Acronyms	**Caption:** PERE = Use of renewable primary energy excluding the renewable primary energy resource used as raw materials; PERM = Use of renewable primary energy as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding the non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste Indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed (HWD)	kg	1.61E-07	0.00E+00	7.65E-12	0.00E+00	5.23E-10	-1.14E-04
Non-hazardous waste disposed (NHWD)	kg	2.17E+01	0.00E+00	1.24E-03	0.00E+00	1.20E+02	1.84E+02
Radioactive waste disposed (RWD)	kg	1.09E-01	0.00E+00	9.80E-06	0.00E+00	2.73E-04	1.67E-03
Materials for Recycling (MFR)	kg	0.00E+00	0.00E+00	0.00E+00	8.80E+02	0.00E+00	0.00E+00
Acronyms	**Caption:** HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Additional environmental performance indicators according to EN 15804+A2:2019

Results per 1 Tonne of Steel Plate							
Impact indicators	**Unit**	**A1-A3**	**C1**	**C2**	**C3**	**C4**	**D**
Particulate matter	Disease incidences	5.69E-04	0.00E+00	7.30E-07	0.00E+00	1.57E-07	-3.51E-05
Ionising radiation, human health	kBq U235 eq.	1.12E+01	0.00E+00	8.91E-04	0.00E+00	3.16E-02	3.43E+01
Ecotoxicity, freshwater	CTUe	8.35E+03	0.00E+00	3.67E+01	0.00E+00	1.51E+01	-7.98E+02
Human toxicity, cancer	CTUh	7.07E-07	0.00E+00	6.00E-10	0.00E+00	2.02E-09	6.24E-07
Human toxicity, non-cancer	CTUh	1.91E-05	0.00E+00	1.29E-08	0.00E+00	2.13E-07	2.98E-06
Land Use	Pt	1.18E+03	0.00E+00	8.76E-02	0.00E+00	5.83E+00	-1.97E+02
Acronyms	**Caption:** PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impacts						

References

- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14. Construction Products, Version 1.3.2
- Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data; CEN/TR 15941:2010
- EN 15804: EN 15804:2012+A2:2019: Sustainability of construction works -Environmental Product Declarations - Core rules for the product category of construction products.
- EN ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- EN ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- EN ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines.
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2022 (https://www.gabi-software.com/support/gabi)